

Banking Customer Churn Analytics

Project Report

1. Problem Statement

Customer churn is a major challenge for banks because losing existing customers can negatively affect revenue, customer relationships, and long-term business growth. The objective of this project is to analyze banking customer data and identify the factors and customer segments associated with higher churn rates.

The project uses customer demographic, financial, product, and activity-related information to understand churn patterns. Power BI is used to transform the data into an interactive analytical dashboard containing KPIs, charts, filters, and business insights. The analysis helps identify high-risk customer segments and provides data-driven insights that can support customer retention strategies.

2. Project Objectives

- Analyze the overall customer churn rate.
- Identify customer segments with higher churn rates.
- Compare churn across different age groups and credit-score groups.
- Analyze the relationship between churn and the number of products held by customers.
- Compare churn rates across different geographical regions and genders.
- Analyze the effect of customer activity status on churn.
- Analyze churn patterns across different balance ranges.
- Create interactive KPI cards and slicers for easy data exploration.
- Develop a professional Power BI dashboard for business analysis.

- Generate actionable insights that can help banks understand customer retention risks.

3. Dataset Description

The Banking Customer Churn dataset contains information about 10,000 bank customers. The dataset includes demographic, financial, product-related, and customer activity information. The main purpose of the dataset is to analyze customer behavior and identify patterns associated with customer churn.

The important attributes used in the analysis are:

- **CustomerID:** Unique identification number assigned to each customer.
- **CreditScore:** Credit score of the customer.
- **Geography:** Country or geographical region of the customer.
- **Gender:** Gender of the customer.
- **Age:** Age of the customer.
- **Tenure:** Number of years the customer has been associated with the bank.
- **Balance:** Account balance maintained by the customer.
- **NumOfProducts:** Number of banking products used by the customer.
- **HasCreditCard:** Indicates whether the customer has a credit card.
- **IsActiveMember:** Indicates whether the customer is an active member of the bank.
- **EstimatedSalary:** Estimated annual salary of the customer.
- **Churn:** Indicates whether the customer has left the bank.

The dataset contains 8,000 retained customers and 2,000 churned customers, resulting in an overall churn rate of 20%.

4. Tools and Technologies

The following tools and technologies were used to develop the Banking Customer Churn Analytics project:

- **Microsoft Power BI:** Used for data visualization, dashboard development, interactive filtering, and business analysis.

- **DAX (Data Analysis Expressions):** Used to create calculated measures and perform churn-related calculations.
- **Power BI Data Modeling:** Used to organize and analyze the customer dataset.
- **Excel/CSV:** Used as the source format for the customer dataset.
- **Data Visualization:** Used to represent customer churn patterns through charts, KPI cards, and interactive dashboards.

5. Data Cleaning and Transformation

The customer dataset was prepared and transformed in Power BI before creating the final dashboard. The following steps were performed:

- Checked the available customer attributes and their data types.
- Verified the customer records and dataset structure.
- Analyzed the Churn column to identify retained and churned customers.
- Created a **Churn Rate** measure to calculate the percentage of customers who churned.
- Created **Age Groups** to analyze churn across different age ranges.
- Created **Credit Score Groups** to analyze churn based on credit-score ranges.
- Created **Balance Groups** to categorize customers according to their account balance.
- Created a **Balance Group Order** column to display balance categories in the correct sequence.
- Analyzed customer churn based on the number of products used.
- Configured interactive slicers for Geography, Gender, Active Membership, and Credit Card ownership.
- Prepared the transformed data for visualization and dashboard analysis.

6. DAX Measures

DAX measures were created in Power BI to calculate important customer churn KPIs.

The major measures used in the project include:

Total Customers

```
Total Customer = COUNTROWS('bank_customer_churn')
```

Churned Customers

```
Churned Customers =  
CALCULATE(  
    COUNTROWS('bank_customer_churn'),  
    'bank_customer_churn'[Churn] = 1  
)
```

Retained Customers

```
Retained Customers =  
CALCULATE(  
    COUNTROWS('bank_customer_churn'),  
    'bank_customer_churn'[Churn] = 0  
)
```

Churn Rate

```
Churn Rate =  
DIVIDE(  
    [Churned Customers],  
    [Total Customer],  
    0  
)
```

The measures were used to create KPI cards and analytical visualizations in the Power BI dashboard.

7. Dashboard Development

The Power BI report was designed using two main dashboard pages.

Page 1 — Banking Customer Churn Analytics

The first page provides an overall view of customer churn performance. It contains:

- Total Customer KPI
- Churned Customer KPI
- Retained Customer KPI
- Overall Churn Rate KPI
- Geography slicer
- Gender slicer
- Active Member slicer
- Credit Card slicer
- Churn Rate by Geography
- Churn Rate by Age Group
- Churn Rate by Credit Score Group
- Churn Rate by Number of Products
- Churn Rate by Balance Group
- Churn Rate by Gender
- Churn Rate by Active Member
- Churn Rate by Credit Card ownership

Page 2 — Customer Churn Insights

The second page focuses on identifying high-risk customer segments and understanding customer behavior.

It contains:

- Churn Rate by Age Group
- Churn Rate by Number of Products
- Churn Rate by Credit Score Group
- Churn Rate by Geography
- Churn Rate by Active Member
- Churn Rate by Gender
- Key Business Insights section

8. Key Business Insights

The dashboard analysis produced several important observations:

- Germany has the highest churn rate among the analyzed geographical regions, at approximately 34.63%.
- Customers in the 66–70 age group show the highest churn rate, at approximately 40.45%.
- Customers with credit scores between 350–499 have a churn rate of approximately 38.29%.
- Inactive members have a significantly higher churn rate than active members.
- Customers using 3–4 banking products show churn rates above 32%.
- The overall customer churn rate in the dataset is 20%.

These insights can help identify customer segments that may require greater attention from the bank's customer retention teams.

9. Conclusion

The Banking Customer Churn Analytics project demonstrates how Power BI can be used to transform customer data into meaningful business insights. The interactive dashboard provides a clear view of overall churn performance and allows users to analyze churn across demographic, financial, geographical, and behavioral dimensions.

The analysis highlights several customer segments associated with higher churn rates, including older customers, inactive members, customers with lower credit scores, and customers using multiple banking products. These findings can help businesses better understand customer behavior and support data-driven customer retention initiatives.

Overall, the project demonstrates practical skills in data analysis, DAX, data visualization, dashboard development, and business intelligence using Microsoft Power BI.